Light

GCSE Computer Science - Ethical, Legal and Environmental Impacts (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is 'software piracy'?

- ○ **A.** Using software on a boat
- ○ **B.** The illegal copying, distribution, or use of software without proper licensing or payment
- ○ **C.** Testing software for bugs
- ○ **D.** Writing software in a foreign country

Q2 (1 mark)

What are 'terms of service' (ToS) when using an app or website?

- ○ **A.** The legal agreement between user and provider that specifies rules, rights, and responsibilities for using the service
- ○ **B.** The speed at which the service operates
- ○ **C.** The opening hours of customer support
- ○ **D.** The colours used in the app design

Q3 (1 mark)

What is a 'data broker'?

- ○ **A.** A person who repairs computers
- ○ **B.** A company that collects personal information about individuals from various sources and sells it to other businesses
- ○ **C.** A type of internet connection
- ○ **D.** Software that speeds up data transfer

Q4 (1 mark)

Some parents use GPS-enabled devices to track their children's location. What is this technology used for?

- ○ **A.** Making children run faster
- ○ **B.** Improving children's school grades
- ○ **C.** Child safety - allowing parents to know their child's location in real-time for peace of mind and emergency response

- ○ **D.** Making phones cheaper for children

Q5 (1 mark)

What security feature helps protect mobile payment systems like Apple Pay or Google Pay?

- ○ **A.** The phone's colour
- ○ **B.** The size of the phone screen
- ○ **C.** The phone's weight
- ○ **D.** Tokenisation - where actual card numbers are replaced with unique codes for each transaction, so real card details are never shared with merchants

Q6 (1 mark)

Some people have expressed health concerns about 5G mobile networks. What do scientific health organisations generally conclude?

- ○ **A.** Current evidence does not show 5G poses health risks at permitted exposure levels, though research continues as with all technologies
- ○ **B.** 5G is definitely dangerous and should be banned
- ○ **C.** 5G has been proven to cure diseases
- ○ **D.** No research has ever been conducted on 5G

Q7 (1 mark)

What is a 'password manager'?

- ○ **A.** A person who remembers passwords for you
- ○ **B.** A website that displays all common passwords
- ○ **C.** Software that securely stores and manages multiple passwords, requiring users to remember only one master password
- ○ **D.** Hardware that types passwords automatically

Q8 (1 mark)

Body cameras worn by police officers or delivery workers are an example of what type of technology?

- ○ **A.** Virtual reality
- ○ **B.** Artificial intelligence
- ○ **C.** Cloud storage
- ○ **D.** Wearable technology that records video evidence of interactions and incidents

Q9 (1 mark)

What does a computerised insulin pump do for people with diabetes?

- ○ **A.** It automatically delivers precise doses of insulin based on programmed settings or sensor readings, helping manage blood sugar levels
- ○ **B.** It helps them see better
- ○ **C.** It improves their hearing

- ○ **D.** It makes their heart beat faster

Q10 (1 mark)

What is an 'autonomous bus'?

- ○ **A.** A bus that is very expensive
- ○ **B.** A bus that only runs on weekends
- ○ **C.** A self-driving bus that can navigate routes and transport passengers without a human driver using sensors and AI
- ○ **D.** A bus painted in bright colours

Q11 (1 mark)

Why are electronic recycling programmes important for the environment?

- ○ **A.** They make electronics run faster
- ○ **B.** They improve internet speeds
- ○ **C.** They make new electronics cheaper
- ○ **D.** They recover valuable materials like gold and copper while safely disposing of hazardous substances like lead and mercury that would otherwise pollute landfills

Q12 (1 mark)

What is 'internet addiction' or 'problematic internet use'?

- ○ **A.** A pattern of excessive internet use that interferes with daily life, relationships, work, or health, where the person struggles to control their usage despite negative consequences
- ○ **B.** Using the internet at work
- ○ **C.** Having a fast internet connection
- ○ **D.** Using the internet to learn new things

Q13 (2 marks)

Many employers monitor employees' computer use, including emails, websites visited, and keystrokes. What are the ethical considerations of workplace digital monitoring?

- ○ **A.** Employers have no right to monitor anything employees do
- ○ **B.** Employees should be monitored 24 hours a day including at home
- ○ **C.** Employers have legitimate interests in productivity, security, and legal compliance, but extensive monitoring can erode trust, invade privacy during personal moments, create stress, and shift workplace culture toward surveillance rather than autonomy
- ○ **D.** Monitoring technology does not actually work

Q14 (2 marks)

Websites use 'cookies' and tracking technologies to show users targeted advertisements based on their browsing history. What privacy concerns does this raise?

- ○ **A.** Targeted advertising has no privacy implications
- ○ **B.** Targeted ads are always more useful than non-targeted ads

○ **C.** Cookies only track which biscuits people like

○ **D.** Tracking creates detailed profiles of users' interests, behaviours, and activities across websites - often without meaningful consent - allowing companies to know intimate details about people's lives, relationships, health concerns, and vulnerabilities

Q15 (2 marks)

A major cloud service provider experiences an outage lasting several hours. What risks does this create for businesses relying on cloud services?

○ **A.** Businesses may be unable to access critical data and applications, halting operations; employees cannot work; customers cannot be served; revenue is lost; and companies have no control over restoration timing - highlighting dependency risks of relying on external providers

○ **B.** Cloud outages have no effect on businesses

○ **C.** Only the cloud provider is affected by outages

○ **D.** Outages make cloud services run faster afterwards

Q16 (2 marks)

What is 'hacktivism' and what ethical questions does it raise?

○ **A.** Hacktivism is a type of exercise programme

○ **B.** Hacktivism refers to very active computer users

○ **C.** Hacktivism is using hacking techniques for political or social activism - raising questions about whether illegal actions can be justified by political goals, and whether hacktivists are freedom fighters, criminals, or vigilantes bypassing democratic processes

○ **D.** Hacktivism is a brand of computer hardware

Q17 (2 marks)

Some payment systems now use fingerprint or facial recognition instead of PINs or signatures. What are the advantages and concerns of biometric payment authentication?

○ **A.** Biometric payments have no advantages or disadvantages

○ **B.** Fingerprints and faces cannot be used for payments

○ **C.** Biometric payments are always slower than typing PINs

○ **D.** Advantages include convenience (nothing to remember or carry) and security (biometrics are harder to steal than passwords); concerns include biometric data being permanent (cannot be changed if compromised), potential for false rejections, and privacy implications of storing sensitive body data

Q18 (2 marks)

What is a 'deepfake' and why is this technology concerning?

○ **A.** Deepfakes are AI-generated fake videos or audio that convincingly show people saying or doing things they never did - concerning because they can spread misinformation, damage reputations, enable fraud, and undermine trust in authentic evidence

○ **B.** A deepfake is a very deep swimming pool

○ **C.** Deepfakes are files that are hidden deep in computer folders

○ **D.** Deepfakes are a type of computer virus

Q19 (2 marks)

Farmers are beginning to use autonomous tractors and drones for agriculture. What benefits and concerns does this raise?

- ○ **A.** Autonomous farming only benefits technology companies
- ○ **B.** Autonomous farming equipment cannot work on farms
- ○ **C.** Benefits include increased efficiency, precision agriculture reducing waste, operation in conditions unsafe for humans, and addressing labour shortages; concerns include high costs, job displacement for farm workers, technology dependence, cybersecurity risks, and questions about rural community impacts
- ○ **D.** All farms already use fully autonomous equipment

Q20 (2 marks)

Smartphones now include 'digital wellbeing' features that show usage statistics and allow setting time limits. What is the purpose and limitation of these tools?

- ○ **A.** Digital wellbeing features make phones run faster
- ○ **B.** Digital wellbeing tools do not exist on smartphones
- ○ **C.** These features permanently delete apps after time limits
- ○ **D.** Purpose is to help users understand and control their technology use; limitations include being easily overridden, not addressing underlying design that makes apps compelling, and potentially transferring responsibility from companies designing addictive products to individual users struggling with them

Q21 (2 marks)

Under data protection laws like GDPR, individuals have a 'right to data portability'. What does this mean and why does it matter?

- ○ **A.** The right to receive personal data you provided to a service in a usable format and transfer it to another provider - important because it reduces lock-in effects that trap users with services holding their data hostage
- ○ **B.** The right to carry your phone anywhere
- ○ **C.** The right to use portable chargers
- ○ **D.** Data portability means data can be carried by ships

Q22 (2 marks)

Some police forces use 'predictive policing' software that uses AI to forecast where crimes are likely to occur. What concerns does this raise?

- ○ **A.** Predictive policing software always prevents all crime
- ○ **B.** Predictive policing has no effect on police decisions
- ○ **C.** Concerns include algorithms trained on biased historical data perpetuating discrimination, creating self-fulfilling prophecies where predicted areas receive more policing which finds more crime, and the ethical implications of treating entire communities as suspects based on statistical patterns rather than individual behaviour
- ○ **D.** These systems predict the future with 100% accuracy

Q23 (2 marks)

Governments are considering requiring age verification for accessing certain online content. What challenges does implementing age verification create?

- ○ **A.** Age verification is simple and has no implementation challenges
- ○ **B.** Challenges include verifying ages reliably without excessive data collection, protecting the privacy of verified users, preventing workarounds, ensuring systems do not exclude legitimate adult users, and balancing child protection against creating surveillance infrastructure or restricting adult freedoms
- ○ **C.** Children cannot access the internet so verification is unnecessary
- ○ **D.** Age verification has been perfectly implemented everywhere already

Q24 (3 marks)

During the shift to remote work, many companies installed monitoring software on employees' home computers that tracks keystrokes, takes screenshots, and monitors application use. Analyse the ethical implications of such surveillance of remote workers.

- ○ **A.** Remote worker monitoring has no ethical implications
- ○ **B.** Employers have interests in ensuring productivity and preventing misuse of paid time; however, home monitoring extends workplace surveillance into private spaces, captures personal and family activities, creates constant stress of being watched in one's own home, raises questions about trust and autonomy, and may violate reasonable expectations that homes are private spaces - requiring careful balancing of employer needs against employee dignity and privacy
- ○ **C.** Employees should have no privacy even in their own homes
- ○ **D.** Monitoring software does not actually track any activity

Q25 (3 marks)

Social media platforms use AI systems to detect and remove harmful content at scale. Analyse the challenges and limitations of AI content moderation.

- ○ **A.** AI enables moderation at scale impossible for humans alone; but AI struggles with context, satire, and cultural nuance; makes errors removing legitimate content or missing violations; cannot consistently apply subjective judgments about what is harmful; raises concerns about opaque decision-making and limited appeal processes; and may suppress marginalized voices whose speech patterns AI misidentifies
- ○ **B.** AI content moderation is perfect and has no limitations
- ○ **C.** Social media platforms do not moderate any content
- ○ **D.** Human moderators can easily review all content without AI assistance

Q26 (3 marks)

Consumer DNA testing services collect genetic information from millions of users. Analyse the privacy implications of sharing genetic data with commercial companies.

- ○ **A.** Genetic data has no privacy implications
- ○ **B.** Genetic information cannot reveal anything meaningful about a person
- ○ **C.** DNA testing companies immediately destroy all samples after testing
- ○ **D.** Genetic data is uniquely sensitive: it is permanent, reveals health predispositions, identifies relatives who never consented, and could be used by insurers, employers, or law enforcement; once shared with a company it may be sold, subpoenaed, hacked, or used for research without meaningful ongoing consent - implications extend far beyond the individual who submitted the sample

Q27 (3 marks)

Schools increasingly use educational technology platforms that collect detailed data about students' learning, behaviour, and performance. Analyse the benefits and concerns of this data collection.

- ○ **A.** Educational technology data collection has only benefits with no concerns
- ○ **B.** Benefits include personalising learning, identifying struggling students early, and enabling evidence-based teaching improvements; concerns include collecting sensitive data about minors who cannot meaningfully consent, creating permanent records that follow students, commercialisation of student data, cybersecurity risks to children's information, and potential for algorithmic tracking to limit students' perceived potential
- ○ **C.** Schools do not use any technology that collects data
- ○ **D.** Student data has no value and is never misused

Q28 (3 marks)

Companies are launching thousands of satellites to provide global internet coverage. Analyse the benefits and environmental concerns of satellite internet constellations.

- ○ **A.** Satellite internet has no environmental impact
- ○ **B.** Satellites provide internet without using any technology
- ○ **C.** Benefits include connecting remote and underserved areas without ground infrastructure; concerns include rocket launch emissions, space debris from thousands of satellites that could trigger collision cascades, interference with astronomical observations, uncertain end-of-life disposal, and the cumulative environmental impact of manufacturing, launching, and replacing satellite constellations
- ○ **D.** Space has unlimited capacity for satellites with no

consequences

Q29 (3 marks)

Large retailers use autonomous robots in warehouses to pick, pack, and transport goods. Analyse the social and economic implications of warehouse automation.

- ○ **A.** Warehouse automation has no social or economic implications
- ○ **B.** Automation increases efficiency and can improve worker safety by handling dangerous tasks; but it displaces workers in one of the largest employment categories, often affecting people with limited alternative options, changes remaining jobs to monitoring machines rather than direct work, raises questions about wealth distribution when productivity gains flow to owners while workers lose livelihoods, and transforms community economies dependent on warehouse employment
- ○ **C.** Warehouse robots only create jobs without eliminating any
- ○ **D.** All warehouse work has already been fully automated

Q30 (3 marks)

Many services are moving online, from banking to government applications to healthcare bookings. Analyse the ethical implications of this digital shift for elderly people who may lack digital skills or access.

- ○ **A.** Moving services online has no implications for elderly people
- ○ **B.** Online services are more difficult for everyone equally
- ○ **C.** All elderly people are expert technology users
- ○ **D.** Digital service migration offers convenience and efficiency for many users; but elderly people may lack digital skills, suitable devices, reliable internet, or confidence with technology - creating exclusion from essential services, increased isolation, loss of independence when

they need help for tasks they once did alone, and ethical questions about whether efficiency gains for the majority justify excluding vulnerable minorities from full participation in society

Total: **55** marks • Time: **30 min**

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